

Exercise No. 5

ASYMMETRIC CRYPTOGRAPHY

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Task 3.1: RSA demonstration tool experiments

- **Public key value (e, n):** $2^{16}+1$
- **Private key value (d, n):** 44273
- **Results:**

Input text

Hello this is my test message

The Input text will be separated into segments of Size 1 (the symbol '#' is used as separator).

H # e # l # l # o # # t # h # i # s # # i # s # # m # y # # t # e # s # t # # m # e # s # s # a # g # e

Numbers input in base 10 format.

072 # 101 # 108 # 108 # 111 # 032 # 116 # 104 # 105 # 115 # 032 # 105 # 115 # 032 # 109 # 121 # 032 #

Encryption into ciphertext $c[i] = m[i]^e \pmod{N}$

31442 # 07428 # 41521 # 41521 # 34310 # 09394 # 10710 # 23366 # 20714 # 06205 # 09394 # 20714 # 01

Task 3.2: RSA encryption with different block lengths

- **Ciphertext (block length 2):** 06415 # 44316 # 16582 # 00095 # 12284 # 27439
- **Ciphertext (block length 1):** 45406 # 108352 # 231578 # 398572 # 323500 # 293215 # 537036 # 490695 # 537036 # 496898 # 340384 # 66484
- **Conclusions:** Encrypting with a block length of 1 character essentially turns RSA into a monoalphabetic substitution cipher, which is insecure and vulnerable to frequency analysis. Using a longer block length increases security by hiding single-character frequencies.

Task 3.3: 1024-bit RSA key pair generation

- **Variables composing the public key:** Modulus (n) and Public Exponent (e)
- **Value of the exponent:** 65537
- **Certificate information:**

Version: 2 (X.509v3-1996)
SubjectName: CN=Student Student [1776865325], DC=cryptool, DC=org
IssuerName: CN=CrypTool CA 2, DC=cryptool, DC=org
SerialNumber: F9:C9:87:CF:F9:2F:E8:85
Validity - NotBefore: Wed Apr 22 15:42:08 2026 (260422134208Z)

NotAfter: Thu Apr 22 15:42:08 2027 (270422134208Z)
Public Key Fingerprint: E1D7 A97B 2AB2 396E 6C90 F86A C315 1F11
SubjectKey: Algorithm rsa (OID 2.5.8.1.1), Keysize = 1024

Public modulus (no. of bits = 1024):

0 FD6E15AE EF7EE66A BCB8B49D FC85E0B8
10 DE978BB8 E46B103F B98E511E 47C40686
20 10D4651F CB1F096B F4819A9D 2C352B40
30 04E5EAC4 D18A121B 92037DE6 6E133833
40 F8897E94 244273F9 98075E25 C5B1D687
50 983FF295 33715CE4 A41B859A 15F7129C
60 D2EF12B0 B680F7A0 A8BC483D 4175A30B
70 803F5172 80710584 ACBC84B8 9DC461D7

Public exponent (no. of bits = 17):

0 010001

Certificate extensions:

Private extensions:

OID 2.206.5.4.3.2:

PrintableString:

|[Student][Student][RSA-1024][177|
|6865325] |

SHA1 digest of DER code of ToBeSigned:

0 BA266B40 7D6D291C 5EBF33DF E27E598A
10 F84AFDD6

Signature: Algorithm sha1WithRSASignature (OID 1.3.14.3.2.29), NULL

0 111E9DC2 F629EC8A E2B56F5A 5631EB8D
10 17A5AD42 3459D709 9527CE18 D63481AB
20 F6ECBC0F A955600D 9DDB92D5 BF017540
30 75578B6C 907216E0 657E5FC4 92DBDBEC
40 810A8F38 56C85F4D C4EEBF5B DF323C92
50 CBF6AF1B B0166967 1D0CB264 E29D18ED
60 4F4B6F2B E530E418 18F1F7A4 3468A1DD
70 6EF19203 6CAE7295 8DC6C77D 5562BC0B
80 F7DA28D4 7E2F382A 95F02AEC C60ECD88
90 776DA44C DAF18CFD E393FA28 BA22E6F4
A0 9527FFB2 401E96AD 3CE69063 92F0EE14
B0 62A3ECD3 F204A07D 93B361A0 9135FEF7
C0 78EF13F7 189CB34C F877050C 95CB20C5
D0 7A8F9D79 0C76103B B4864826 4B016F70
E0 FD8CAEF7 B09DEDD9 9D6F9A47 359034DE
F0 BB07CCC3 DFD608A9 7FAD0854 5043C68E

Certificate Fingerprint (MD5): 77:F6:BF:C8:34:DB:4D:27:DA:93:DE:9C:80:96:D7:C5

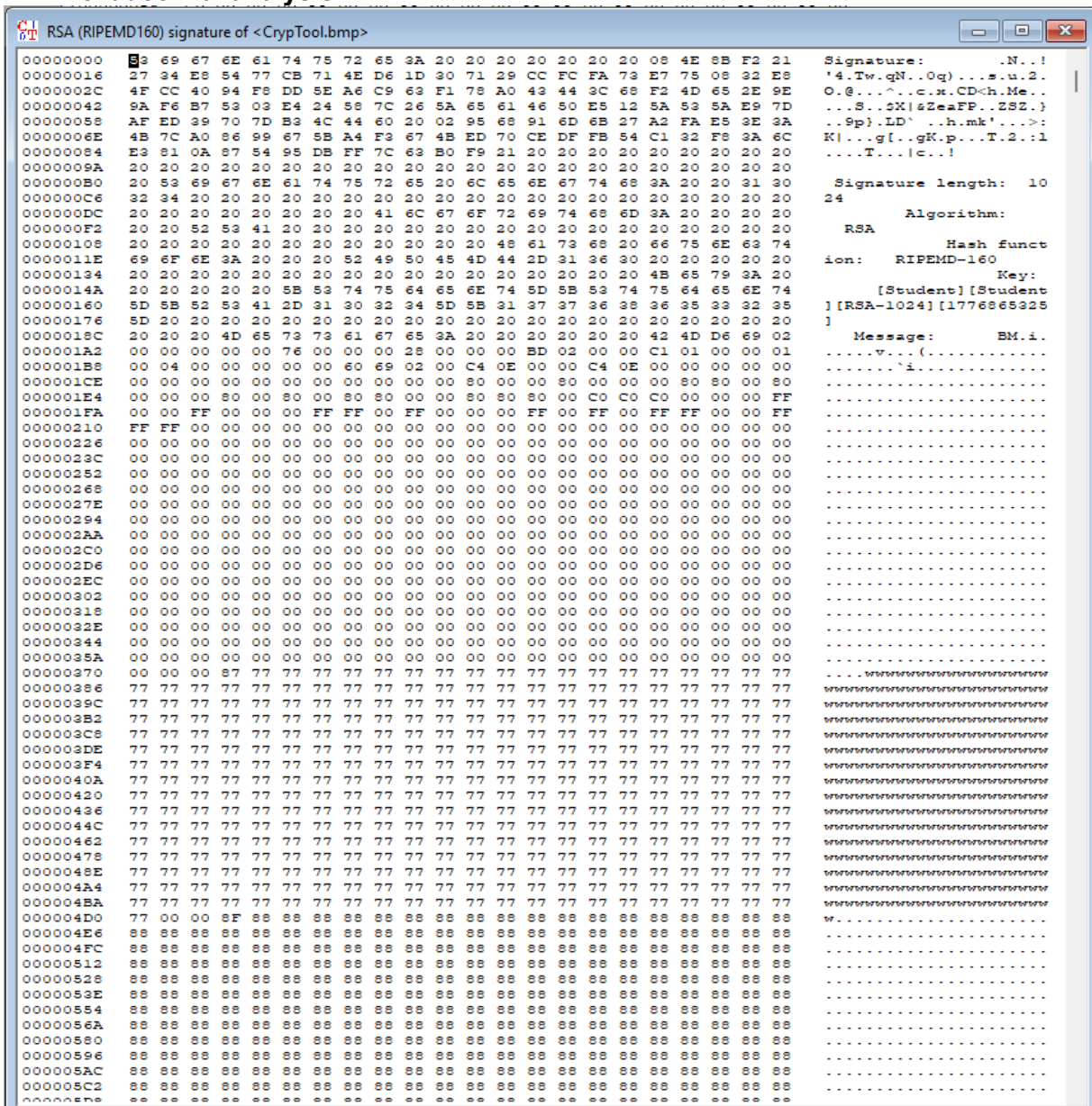
Certificate Fingerprint (SHA-1): 898F E19E CD15 865A 0858 17D4 6F45 0B88 635B F3C3

Task 3.4: RSA encryption and decryption duration

- **Encryption duration:** 0.000s
- **Decryption duration:** 0.014s
- **Conclusions:** Decryption typically takes significantly longer than encryption in RSA. This is because the private exponent (d) is usually much larger than the public exponent (e), requiring more computational time for the modular exponentiation process.

Task 3.5: Digital signature embedding

- **Duration of signing:** 0.004s
- **Hexadecimal analysis:**



- **Is it possible to separate signature and data?:** Yes, the signature is typically appended or prepended to the data file. In hexadecimal view, the signature block can be identified and separated from the original file data.

Task 3.6: Impact of errors on the digital signature

- **Signature verification before changing:** Verification succeeded.
- **Signature verification after changing signature characters:** Verification fails. The signature is recognized as invalid.

- ### Task 3.7: Factorization of module n and calculating d

- ### Task 3.8: RSA Cryptoanalysis using factorization

RSA Demonstration

RSA using the private and public key -- or using only the public key

- ☒ Choose two prime numbers p and q. The composite number N = pq is the public RSA modulus, and $\phi(N) = (p-1)(q-1)$ is the Euler totient. The public key e is freely chosen but must be coprime to the totient. The private key d is then calculated such that $d = e^{-1} \pmod{\phi(N)}$.
- ☐ For data encryption or certificate verification, you will only need the public RSA parameters: the modulus N and the public key e.

Prime number entry

Prime number p	145295143558111	Generate prime numbers...
Prime number q	440334654777631	

RSA parameters

RSA modulus N	63978486879527143858831415041	(public)
$\phi(N) = (p-1)(q-1)$	63978486879526558229033079300	(secret)
Public key e	17579	
Private key d	10663687727232084624328285019	

RSA encryption using e / decryption using d [alphabet size: 27]

Input as ☐ text ☒ numbers Alphabet and number system options...

Ciphertext coded in numbers of base 10

7091621432020076311552201 # 46468979279750354732637631044 # 32870167545903741339819671379

Decryption into plaintext $m[i] = c[i]^d \pmod{N}$

00000000000001401202118011200 # 00000000000001421130205181900 # 000000000000011805001301

Output text from the decryption (into segments of size 14; the symbol '#' is used as separator).

NATURAL # NUMBERS # ARE MADE # BY GOD

Plaintext

NATURAL NUMBERS ARE MADE BY GOD

Encrypt Decrypt Close

Task 3.9: Attack on short private key

- **p:** 17827213697419331357
- **q:** 10191818653674809381
- **Delta coefficient:** 0,1999
- **Check on numbers from Task 3.1:** The attack is only mountable for $\delta < 0,29$

Task 3.10: Advantages and disadvantages of asymmetric cryptography

- **Advantages:**
 - **Key Distribution:** Eliminates the need for secure exchange of a secret key, solving the primary key distribution problem of symmetric cryptography.
 - **Digital Signatures:** Enables non-repudiation and authentication, allowing users to verify the sender's identity and data integrity.
 - **Scalability:** Requires fewer keys overall for a network of users compared to symmetric systems (n pairs instead of $n(n-1)/2$ keys).
- **Disadvantages:**
 - **Computational Cost:** Significantly slower and more computationally intensive than symmetric cryptography due to complex mathematics involving very large numbers.
 - **Key Size:** Requires much larger key sizes (e.g., 2048-bit or 4096-bit) to achieve the same level of security as symmetric algorithms (e.g., 128-bit or 256-bit).
 - **Vulnerability to Quantum Computing:** Algorithms like RSA and ECC are theoretically vulnerable to Shor's algorithm and could be broken by a sufficiently powerful quantum computer.